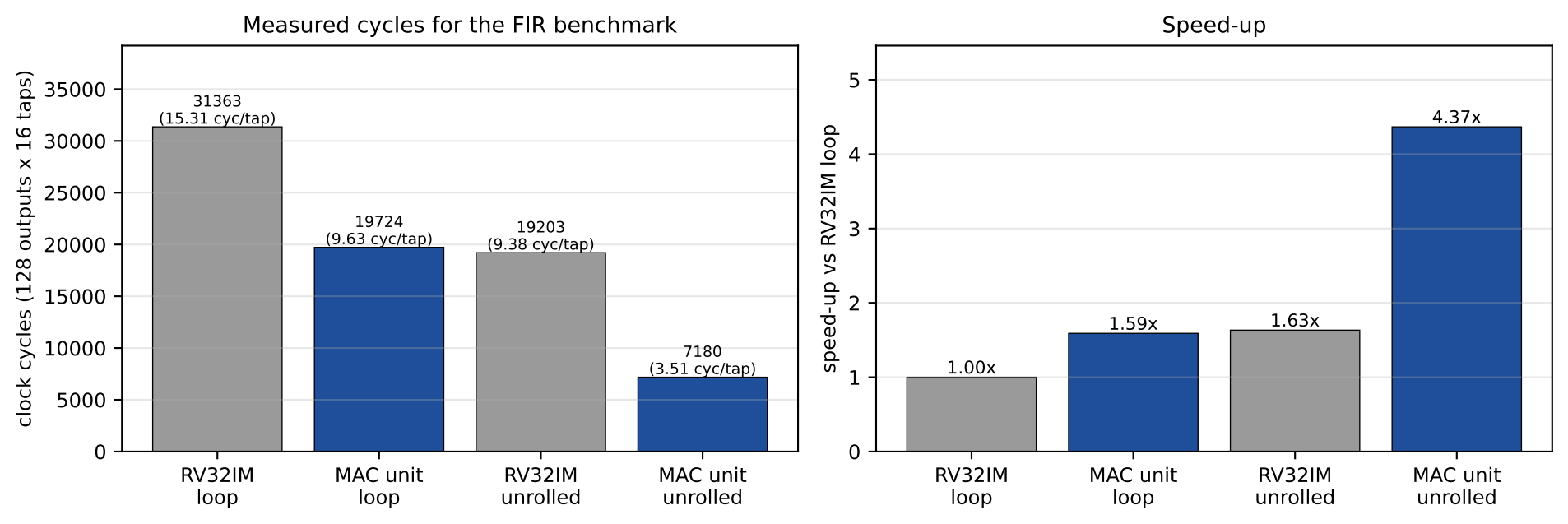
Measured cycles for the FIR benchmark
clock cycles (128 outputs x 16 taps)
0
5000
10000
15000
20000
25000
30000
35000
31363
(15.31 cyc/tap)
19724
(9.63 cyc/tap)
19203
(9.38 cyc/tap)
7180
(3.51 cyc/tap)
RV32IM loop
MAC unit loop
RV32IM unrolled
MAC unit unrolled
Speed-up
speed-up vs RV32IM loop
0
1
2
3
4
5
1.00x
1.59x
1.63x
4.37x
RV32IM loop
MAC unit loop
RV32IM unrolled
MAC unit unrolled